

COURSE CODE/TITLE:	CPE 212	SECTION:	CPE31S2
DATE PERFORMED:	Aug 13, 2026	DATE SUBMITTED:	Aug 13, 2026
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ACTIVITY NO. ____	Prelim Exam Skills
1. PROCEDURE OUTPUT	

Hostname	Operating System	IP Address
server1	Ubuntu	192.168.56.110
server2	Ubuntu	192.168.56.113
server3	CentOS	192.168.56.114

Part A: Installing and Verifying Ansible

Submit a screenshot of the following:

- Ansible version

```
mdmlot@workstation:~$ ansible --version
ansible 2.10.8
  config file = None
  configured module search path = ['/home/mdmlot/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/lib/python3/dist-packages/ansible
  executable location = /usr/bin/ansible
  python version = 3.10.12 (main, Jun 22 2026, 18:55:27) [GCC 11.4.0]
mdmlot@workstation:~$
```

- SSH connectivity

```
mdmlot@workstation:~$ ssh mdmlot@server1
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

70 updates can be applied immediately.
70 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Aug 13 07:58:30 2026 from 192.168.56.112
mdmlot@server1:~$
```

```
mdmlot@workstation:~$ ssh mdmlot@server2
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-134-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

Expanded Security Maintenance for Applications is not enabled.

70 updates can be applied immediately.
70 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Thu Aug 13 07:58:23 2026 from 192.168.56.112
mdmlot@server2:~$
```

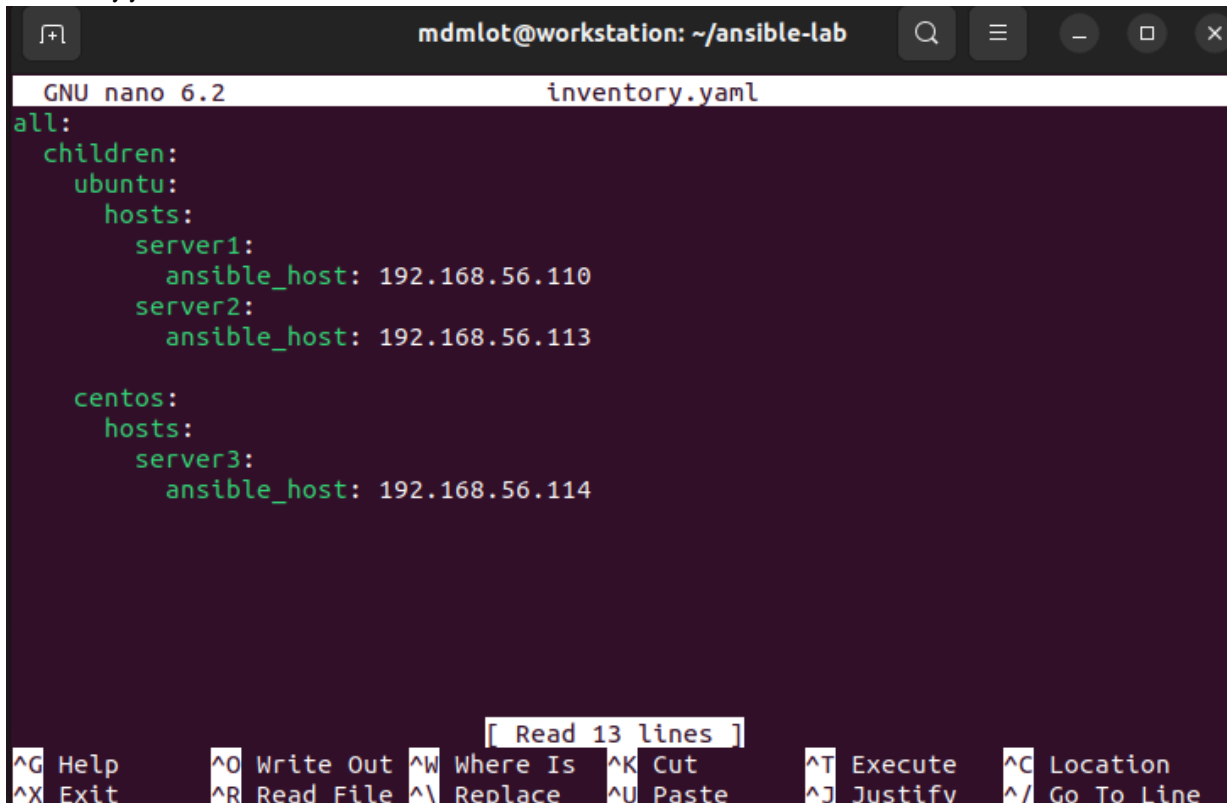
```
mdmlot@workstation:~$ ssh mdmlot@server3
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Aug 13 08:13:04 2026 from 192.168.56.112
[mdmlot@server3 ~]$
```

Part B: Building an Inventory File

Demonstrate the following:

- inventory.yaml



```
mdmlot@workstation: ~/ansible-lab
GNU nano 6.2 inventory.yaml
all:
  children:
    ubuntu:
      hosts:
        server1:
          ansible_host: 192.168.56.110
        server2:
          ansible_host: 192.168.56.113

    centos:
      hosts:
        server3:
          ansible_host: 192.168.56.114

[ Read 13 lines ]
^G Help      ^O Write Out ^W Where Is  ^K Cut      ^T Execute  ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste    ^J Justify  ^/ Go To Line
```

- Successful inventory verification

```
mdmlot@workstation:~/ansible-lab$ ansible-inventory -i inventory.yaml --list
{
  "_meta": {
    "hostvars": {
      "server1": {
        "ansible_host": "192.168.56.110"
      },
      "server2": {
        "ansible_host": "192.168.56.113"
      },
      "server3": {
        "ansible_host": "192.168.56.114"
      }
    }
  },
  "all": {
    "children": [
      "centos",
      "ubuntu",
      "ungrouped"
    ]
  },
  "centos": {
    "hosts": [
      "server3"
    ]
  },
  "ubuntu": {
    "hosts": [
      "server1",
      "server2"
    ]
  }
}
```

```
mdmlot@workstation:~/ansible-lab$ ansible all --list-hosts
hosts (3):
  192.168.56.110
  192.168.56.113
  192.168.56.114
mdmlot@workstation:~/ansible-lab$
```

- Successful ping results

```
mdm1ot@workstation:~/ansible-lab$ ansible all -m ping -i inventory.yaml
server1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
server2 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward compatibility with
prior Ansible releases. A future Ansible release will default to using the discovered
platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for
more information. This feature will be removed in version 2.12. Deprecation warnings can be
disabled by setting deprecation_warnings=False in ansible.cfg.
server3 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python"
  },
  "changed": false,
  "ping": "pong"
}
```

Part C: Creating a Metagroup

```
mdm1ot@workstation:~/ansible-lab$ ansible linuxservers -m ping -i inventory.yaml
server2 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
server1 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python3"
  },
  "changed": false,
  "ping": "pong"
}
[DEPRECATION WARNING]: Distribution centos 9 on host server3 should use
/usr/libexec/platform-python, but is using /usr/bin/python for backward compatibility with
prior Ansible releases. A future Ansible release will default to using the discovered
platform python for this host. See
https://docs.ansible.com/ansible/2.10/reference_appendices/interpreter_discovery.html for
more information. This feature will be removed in version 2.12. Deprecation warnings can be
disabled by setting deprecation_warnings=False in ansible.cfg.
server3 | SUCCESS => {
  "ansible_facts": {
    "discovered_interpreter_python": "/usr/bin/python"
  },
  "changed": false,
  "ping": "pong"
}
```

```
mdmlot@workstation:~/ansible-lab$ ansible linuxservers --list-hosts -i inventory.yaml
hosts (3):
  server1
  server2
  server3
```

Part D: Creating Variables

```
mdmlot@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml testvars.yaml

PLAY [Display Variables] *****

TASK [Gathering Facts] *****
ok: [server1]
ok: [server2]
ok: [server3]

TASK [Show Operating System] *****
ok: [server1] => {
  "msg": "Host server1 is running Ubuntu"
}
ok: [server2] => {
  "msg": "Host server2 is running Ubuntu"
}
ok: [server3] => {
  "msg": "Host server3 is running CentOS"
}

PLAY RECAP *****
server1      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0
server2      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0
server3      : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0
```

Part E: Creating Your First Playbook

```

admin@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml playbook.yaml

PLAY [First Ansible Playbook] *****

TASK [Gathering Facts] *****
ok: [server2]
ok: [server1]
ok: [server3]

TASK [Ping all managed nodes] *****
ok: [server1]
ok: [server2]
ok: [server3]

TASK [Display Hello World] *****
ok: [server1] => {
  "msg": "Hello World from server1"
}
ok: [server2] => {
  "msg": "Hello World from server2"
}
ok: [server3] => {
  "msg": "Hello World from server3"
}

TASK [Display Operating System] *****
ok: [server1] => {
  "msg": "Operating System: Ubuntu "
}
ok: [server2] => {
  "msg": "Operating System: Ubuntu "
}
ok: [server3] => {
  "msg": "Operating System: CentOS "
}

PLAY RECAP *****
server1      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0
server2      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0
server3      : ok=4    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0
  
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Create a new group named servers that contains only:

- ubuntu1
- centos1

```

centos:
  servers:
  hosts:

server1:
server3:
  
```

Create a playbook named challenge.yaml that displays the message:
Welcome to Ansible Laboratory

```
---
- name: Prelim Supplementary
  hosts: servers

  tasks:
    - name: Display Challenge Text
      debug:
        msg: "Welcome to Ansible Laboratory"
```

Run the playbook and demonstrate that only hosts in the servers group receive the message.

```
mdm1ot@workstation:~/ansible-lab$ ansible-playbook -i inventory.yaml challenge.yaml

PLAY [Prelim Supplementary] *****

TASK [Gathering Facts] *****
ok: [server1]
ok: [server3]

TASK [Display Challenge Text] *****
ok: [server1] => {
  "msg": "Welcome to Ansible Laboratory"
}
ok: [server3] => {
  "msg": "Welcome to Ansible Laboratory"
}

PLAY RECAP *****
server1                : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0
server3                : ok=2    changed=0    unreachable=0    failed=0    skipped=0    rescued=0
  ignored=0
```

Guide Questions

1. What is the role of the Control Node in Ansible?

The role of the Control Node in Ansible is, based from the name itself, to control the Managed Nodes or remote nodes by setting up the inventory.yaml, ansible.cfg, hosts, and other files along with the actual playbooks to run the commands on these remote hosts.

2. Why is an inventory file required?

An inventory file is required to set up, group, and organize the list of managed nodes and their respective IP addresses which acts as a basis on where the playbooks will run their commands.

3. What is the purpose of a metagroup?

Metagroups allow you to organize and create subgroups within groups using the children: keyword, allowing for multiple layers of organization and specification.

4. How do variables simplify Ansible administration?

Variables simplify Ansible administration by enabling code reusability across different environments. Changing values of a variable specified in one specified location will update configurations globally or wherever the variable is set.

5. What is the difference between an ad hoc command and a playbook?

Playbooks act as reusable tasks to be run across a specified system while ad hoc commands are quick and temporary single-line commands for tasks to only be done once.

Conclusion:

I conclude that mastery of configuring the inventory.yaml and the playbook yaml files are important in effective system administration across multiple remote hosts or servers. Defining and organizing groups in the inventory and knowing how to run commands in the playbook yaml files will allow us to significantly speed up processes and save a lot of time over physically setting up each remote host one by one.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

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